NTIRE 2026 Image Denoising ($\sigma = 50$) Challenge Factsheet SRX

Naitik Pal
Indian Institute of Information Technology Una
Himachal Pradesh, India
naitikpal513@gmail.com

Ujjwal Mishra
Indian Institute of Information Technology Una
Himachal Pradesh, India
ujjwalmishra238@gmail.com

Kailash A. Hambarde
University of Beira Interior, Instituto de Telecomunicações
Covilhã, Portugal
kailash.hambarde@ubi.pt

Hugo Proença
University of Beira Interior, Instituto de Telecomunicações
Covilhã, Portugal
hugomcp@ubi.pt

Praful Hambarde
Indian Institute of Technology Mandi
Himachal Pradesh, India
praful@iitmandi.ac.in

Amit Shukla
Indian Institute of Technology Mandi
Himachal Pradesh, India
amitshukla@iitmandi.ac.in

1. Team Details

- **Team name:** Noice
- **Team leader:** Praful Hambarde
- **Team leader address:** Indian Institute of Technology Mandi, Himachal Pradesh, India
- **Team leader phone:** +91 96237 31283
- **Team leader email:** praful@iitmandi.ac.in
- **Team members:** Naitik Pal, Ujjwal Mishra, Kailash A. Hambarde, Hugo Proença, Amit Shukla
- **Team website:** N/A
- **Affiliations:**
  - Indian Institute of Technology Mandi, Himachal Pradesh, India
  - Indian Institute of Information Technology Una, Himachal Pradesh, India
  - University of Beira Interior & Instituto de Telecomunicações, Covilhã, Portugal
- **CodaBench username:** naitikpal
- **Development phase entry ID:** 599941
- **Validation phase entry ID:** 625178
- **Code repository:** https://github.com/VectorCipher/NTIRE2026_Dn50_challenge

2. Method Details

2.1. General Overview

We present **SRX**, our proposed pipeline for the NTIRE 2026 Image Denoising ($\sigma = 50$) challenge. SRX is built upon an ensemble of three state-of-the-art transformer- and CNN-based image restoration networks: Restormer [4], XFormer [5], and SCUNet [6]. Each model is independently fine-tuned on a combined high-quality dataset and subsequently fused via a weighted ensemble strategy at inference time. Test-time augmentation (TTA) with tiled inference is further employed to boost performance and handle high-resolution inputs efficiently. An overview of the SRX pipeline is illustrated in Figure 1.

2.2. Architecture Details

Restormer. Restormer [4] is a hierarchical encoder-decoder transformer that applies multi-head self-attention in a transposed manner across channels rather than spatial dimensions, making it computationally tractable for high-resolution images. We use the standard configuration with `dim` $=$ 48, `num_blocks`$=$ $[4, 6, 6, 8]$, `num_refinement_blocks`$=$ 4, `heads`$=$ $[1, 2, 4, 8]$,

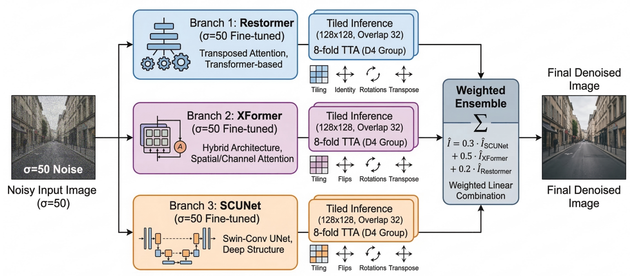

Figure 1. Overview of the proposed **SRX** pipeline.

FFN expansion factor of 2.66, and `BiasFree` layer normalisation (as required for the Gaussian denoising variant). The model is initialised from the official pre-trained checkpoint `gaussian_color_denoising_sigma50.pth` [4].

XFormer. XFormer [5] combines spatial window-based self-attention and channel-wise attention in a unified architecture, giving complementary receptive fields to Restormer. We configure it with `dim` $= 48$, `num_blocks`$=[2, 4, 4]$, `spatial_num_blocks`$= [2, 4, 4, 6]$, `num_refinement_blocks`$= 4$, `heads`$= [1, 2, 4, 8]$, and window size 16×16 at each scale. The model is initialised from its publicly released denoising weights and subsequently fine-tuned on our combined dataset.

SCUNet. SCUNet [6] is a Swin-Conv UNet that integrates Swin Transformer blocks and convolutional residual blocks within a symmetric encoder-decoder structure. We use `config` $= [4, 4, 4, 4, 4, 4, 4]$ (seven-stage depth) and `dim` $= 64$. The model is initialised from the official `scunet_color_50.pth` checkpoint [6] targeting $\sigma = 50$ Gaussian colour noise, and then further fine-tuned.

2.3. Training Data

All three models are fine-tuned on a combined dataset of high-quality clean images:

- **LSDIR** [2]: 2,000 images sampled from the Large Scale Dataset for Image Restoration.
- **DIV2K** [1]: 800 high-resolution training images from the DIV2K dataset.

Gaussian noise with $\sigma = 50$ is synthesised on-the-fly during training, following standard self-supervised Gaussian denoising protocols. No external noisy images are used; ground-truth clean patches serve as targets. No additional data beyond the above two datasets is used.

2.4. Data Augmentation

During training, the following augmentation pipeline is applied to each sampled patch:

- **Random crop:** Patches of size 64×64 are randomly cropped from each image. If the image is smaller than the crop size, it is upscaled to 64×64 via bilinear interpolation.
- **On-the-fly noise synthesis:** Additive white Gaussian noise with $\sigma = 50$ is added to each clean crop, producing the corresponding noisy input.
- **Horizontal and vertical flips:** Random left-right and top-bottom flips are applied.
- **Random rotations:** Random 90° rotations are applied for orientation diversity.

2.5. Training Strategy

Each model is fine-tuned independently using the following protocol.

Loss function. We use the ℓ_1 (mean absolute error) loss between the network output and the clean ground-truth patch:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^{N} \|\hat{x}_i - x_i\|_1, \tag{1}$$

where $\hat{x}_i$ is the restored patch and x_i is the clean reference.

Optimiser. AdamW [3] with an initial learning rate of 1×10^{-4} and default momentum parameters ($\beta_1 = 0.9$, $\beta_2 = 0.999$).

Batch and patch size. Batch size of 4 and patch size of 64×64.

Epochs. 40 epochs for each model.

Mixed precision. Training uses PyTorch Automatic Mixed Precision (AMP) with a `GradScaler` to reduce GPU memory consumption and accelerate training.

Partial freezing (Restormer only). For Restormer, the three encoder levels (`encoder_level1`, `encoder_level2`, `encoder_level3`) are frozen during fine-tuning. Only the bottleneck, decoder, and refinement blocks are updated. This prevents overfitting the low-level feature extraction layers that are already well trained from the original pre-training, while allowing the higher-level reconstruction components to adapt to the challenge data distribution.

2.6. Inference: Tiled Processing

Because the images may be arbitrarily large, direct full-image inference is impractical. We employ overlapping tile-based inference:

- **Tile size:** 128×128 pixels.
- **Overlap:** 32 pixels on each border (stride = $128 - 32 = 96$).
- Overlapping regions are accumulated and averaged by a pixel-wise weight map, eliminating seam artefacts at tile boundaries.

2.7. Test-Time Augmentation (TTA)

To further improve prediction quality, we apply an 8-fold geometric TTA covering all symmetries of the dihedral group D_4:

1. Original (identity)
2. Horizontal flip
3. Vertical flip
4. Horizontal + Vertical flip (180° rotation)
5. Transpose
6. Transpose + Horizontal flip
7. Transpose + Vertical flip
8. Transpose + Horizontal + Vertical flip

Each augmented version of the input is passed through the model (with tiled inference), the inverse geometric transformation is applied to the output, and all eight results are averaged pixel-wise.

2.8. Ensemble

The final denoised image is produced by a weighted linear combination of the outputs of all three fine-tuned models after TTA:

$$\hat{I} = 0.3 \cdot \hat{I}_{\text{SCUNet}} + 0.5 \cdot \hat{I}_{\text{XFormer}} + 0.2 \cdot \hat{I}_{\text{Restormer}}, \quad (2)$$

where $\hat{I}_{\text{SCUNet}}$, $\hat{I}_{\text{XFormer}}$, and $\hat{I}_{\text{Restormer}}$ are the TTA-averaged outputs of SCUNet, XFormer, and Restormer, respectively. The weights were determined empirically on the CodaBench validation leaderboard. XFormer receives the highest weight as it demonstrated the strongest individual performance on the validation split.

2.9. Implementation Details

All experiments are conducted in PyTorch. Training is performed on a single GPU. Inference uses `torch.no_grad()` and mixed-precision autocast. The code, trained model checkpoints, and inference scripts are publicly available at: https://github.com/VectorCipher/NTIRE2026_Dn50_challenge.

2.10. Experimental Results

Our best submission during the development phase (entry ID: 599941) and validation phase (entry ID: 625178) were evaluated on the NTIRE 2026 CodaBench platform. Quantitative PSNR/SSIM results are reported on the official leaderboard.

3. Other Details

- **Paper submission:** We do not plan to submit a solution description paper to the NTIRE 2026 workshop.
- **General impressions:** The NTIRE 2026 challenge provides a well-structured and fair evaluation environment. The CodaBench platform allows rapid iteration and objective comparison across teams. The fixed noise level ($\sigma = 50$) enables focused model development targeting a specific and practically relevant noise regime.
- **Future challenge expectations:** We would welcome tracks covering blind noise estimation (unknown σ), real-world sensor noise, and mixed degradation scenarios. Evaluation metrics beyond PSNR—such as LPIPS or perceptual quality measures—would also enrich the comparison.
- **Difficulties encountered:** Tiled inference with overlapping windows required careful border handling to avoid visible seam artefacts. Partial freezing of Restormer encoder layers was critical to prevent overfitting given the relatively small fine-tuning dataset. Determining the ensemble weights required multiple validation submissions.

References

[1] Eirikur Agustsson and Radu Timofte. Ntire 2017 challenge on single image super-resolution: Dataset and study. In *2017 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 1122–1131, 2017. 2

[2] Yawei Li, Kai Zhang, Jingyun Liang, Jiezhang Cao, Ce Liu, Rui Gong, Yulun Zhang, Hao Tang, Yun Liu, Denis Demandolx, Rakesh Ranjan, Radu Timofte, and Luc Van Gool. Lsdir: A large scale dataset for image restoration. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 1775–1787, 2023. 2

[3] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019. 2

[4] Syed Waqas Zamir, Aditya Arora, Salman Khan, Munawar Hayat, Fahad Shahbaz Khan, and Ming-Hsuan Yang. Restormer: Efficient transformer for high-resolution image restoration. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 5728–5739, 2022. 1, 2

[5] Jiale Zhang, Yulun Zhang, Jinjin Gu, Jiahua Dong, Linghe Kong, and Xiaokang Yang. Xformer: Hybrid x-shaped transformer for image denoising, 2024. 1, 2

[6] Kai Zhang, Yawei Li, Jingyun Liang, Jiezhang Cao, Yulun Zhang, Hao Tang, Deng-Ping Fan, Radu Timofte, and Luc Van Gool. Practical blind image denoising via swin-conv-unet and data synthesis. *Machine Intelligence Research*, 20(6):822–836, 2023. 1, 2